

First steps

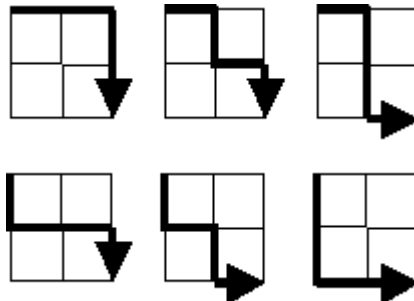
Your high school friend writes to you that they have a client who needs some software developed, they pay well but they only work with companies who work in C# and .NET. You quickly google creating a company in Romania, then C# and .NET both should be easy. Only question is who should be in the team?

Install Visual Studio Community Edition, selecting the following:

- .NET desktop development
- WinUI application development

Form pairs of 2, pick a problem from below and try to solve it while alternating between driver and navigator roles. The driver writes code, while the navigator guides what to write.

1. Starting in the top left corner of a 2 x 2 grid and only being able to move to the right and down, there are exactly 6 routes to the bottom right corner.



How many such routes are there through a 20 x 20 grid? (137846528820 to check)

2. If the numbers 1 to 5 are written out in words: one, two, three, four, five, then there are $3 + 3 + 5 + 4 + 4 = 19$ letters used in total. If all the numbers from 1 to 1000 (one thousand) inclusive were written out in words, how many letters would be used?

NOTE: Do not count spaces or hyphens. For example, 342 (three hundred and forty-two) contains 23 letters and 115 (one hundred and fifteen) contains 20 letters. The use of "and" when writing out numbers follows the British usage. (21124 to check)

3. $10! = 10 * 9 * 8 * 7 * 6 * 5 * 4 * 3 * 2 * 1 = 3628800$, and the sum of the digits in the number $10!$ is $3 + 6 + 2 + 8 + 8 + 0 + 0 = 27$.

Find the sum of the digits in the number $100!$ (648 to check)

4. A permutation is an ordered arrangement of objects. For example, 3124 is one possible permutation of the digits 1, 2, 3 and 4. If all of the permutations are listed numerically or alphabetically, we call it lexicographic order. The lexicographic permutations of 0, 1 and 2 are: 012 021 102 120 201 210

What is the millionth lexicographic permutation of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9? (2783915460 to check)

5. Starting with the number 1 and moving to the right in a clockwise direction a 5 by 5 spiral is formed as follows:

21	22	23	24	25
20	7	8	9	10
19	6	1	2	11
18	5	4	3	12
17	16	15	14	13

It can be verified that the sum of the numbers on the diagonals is 101. What is the sum of the numbers on the diagonals in a 1001 by 1001 spiral formed in the same way? (669171001 to check)

6. The decimal number, $585_{10} = 1001001001_2$ (binary), is palindromic in both bases.

Find the sum of all numbers, less than one million, which are palindromic in base 10 and base 2. (Please note that the palindromic number, in either base, may not include leading zeros.) (872187 to check)

7. Build a sudoku solver that uses deduction, aka do not use brute force to find a solution.